

Future-Splitting State Recruitment

A Cross-Domain Assay for State Adequacy
and Prospective Mechanism Activation

Mark Charles McLaughlin

Independent Researcher

ORCID: 0009-0005-7736-1511

Framework lineage: MKUFT / Addressed Admissible Futures / Typed Viability Traversal

Status. This theory-and-experiment preprint introduces Future-Splitting State Recruitment (FSSR), a falsifiable assay for testing whether a declared state representation remains sufficient for a specified future. Addressed Admissible Futures (AAF) supplies the future-equivalence criterion; FSSR applies that criterion to measured physical trajectories; and the observation/readout construction fixes the target, horizon, challenge family, resolution and tolerance. A state-recruitment event is credited only when a previously adequate state fails under an independently scored future challenge, a minimal physically typed augmentation restores closure on held-out data, and remove/restore tests reproduce the deformation. The principal prospective prediction is that, in systems where an emerging mechanism becomes future-discriminating before its conventional macroscopic marker is overt, recruitment can occur first and can identify the missing mechanism-relevant information. The paper closes the claim into a three-domain experimental triangle: holography as deterministic physical calibration, single-cell dynamics as an independent natural-system anchor, and history-dependent constitutive mechanics as the prospective flagship.

Persistent identifier. Zenodo draft record 22058303. DOI pending publication; no DOI is asserted in this manuscript.

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Compressed statement. A state is adequate only if histories it calls “the same” cannot be forced into materially different declared futures. When a changing system makes such a split possible, the old state has failed. The smallest physically typed information that repairs the split is newly load-bearing for that future. Track the first such recruitment event and ask whether it precedes the visible transition.

Abstract

History-dependent dynamical systems are commonly modelled by augmenting observable state with latent or internal variables. Modern methods can learn such variables, select latent dimension, impose thermodynamic structure, and optimise experiments for parameter identifiability; predictive-state formalisms define state by the futures a history supports under admissible inputs; and active-probing methods can expose approaching transitions. These results leave a narrower empirical question: can the *failure and repair of a state representation itself* serve as a prospective physical signal and mechanism-localisation instrument?

Future-Splitting State Recruitment (FSSR) provides a direct test. A candidate state map Θ is sufficient for target q , horizon Δ , regime λ , and challenge family $\mathcal{U}$ only when histories mapped to the same state remain future-equivalent within a preregistered tolerance. A future-splitting challenge is an admissible continuation chosen to maximise divergence between futures of histories that Θ has merged. If confirmatory divergence exceeds tolerance under an independent continuation source, Θ has failed. The state is then augmented by the smallest preregistered, physically typed coordinate or coalition that restores predictive closure on held-out data; reproducible failure, restoration, and remove/restore deformation define a *state-recruitment event*.

FSSR makes three discriminating predictions. First, where an emerging physical mechanism becomes future-discriminating before its conventional macroscopic marker is overt, recruitment can precede that marker. Second, recruitment can occur without a detectable change in snapshot intrinsic dimension because the missing distinction may reside in history, phase, boundary relation, or another coordinate absent from the instantaneous observable manifold. Third, loading selected to falsify state adequacy need not coincide with loading selected to maximise parameter information. The paper specifies independent-continuation, near-state matching, probe-induction and nested-confirmation gates; a deterministic holographic calibration; a single-cell biological anchor; and a flagship test in learned history-dependent constitutive mechanics. The construction fails if strong ordinary baselines reproduce the same held-out prediction and mechanism localisation without the state-recruitment object.

Keywords: history dependence; internal variables; predictive state; constitutive modelling; state sufficiency; experimental design; active probing; early warning; mechanism activation; multiscale mechanics; readdressing.

1. Scientific gap and contribution boundary

The broad problem is not new. A reduced description can forget information that later matters. In projection-based dynamics, unresolved degrees of freedom can return as memory and noise [1]. In computational mechanics, histories are grouped by equality of predictive futures, producing minimal predictive causal states [2]. Predictive state representations (PSRs) use action-conditional predictions of observable tests as state variables [3, 4]. Input-output computational mechanics extends the same logic to processes driven by inputs [5]. State-identification theory has long used distinguishing input sequences to separate hidden states [6].

Mechanics has independently reached the same seam from the constitutive side. Liu et al. showed that homogenisation can create complex macroscopic history dependence and introduced recurrent neural operators (RNOs) that learn internal variables rather than prescribe them [7]. Their paper explicitly notes that no established method identifies the macroscopic internal variables or even their dimensionality in the general case. Subsequent work has advanced quickly. Temperature-aware RNOs show that non-isothermal load paths can require additional learned state variables [8]; thermodynamics-constrained architectures can recover interpretable path-dependent internal variables under hard physical constraints [9]; and a recent causal energetic formulation establishes identifiability of learned internal variables only up to a linear transformation under controllability assumptions [10]. Bayesian optimal experimental design can now optimise specimen geometry and loading history to improve identification of history-dependent constitutive parameters [11].

Nor is “use a perturbation to see a transition coming” new. Critical slowing down, perturbation response,

functional observability, intrinsic-dimension changes, and active probing can all reveal approaching transitions or their embedding [12–15]. Adaptive latent models can also discover an appropriate latent dimension without fixing it in advance [16].

The present proposal therefore does *not* claim novelty for memory, hidden variables, future-defined state, state identification, adaptive latent dimension, optimal loading, active probing, or early warning separately.

The surviving question is narrower:

Can a changing physical system be interrogated by futures chosen specifically to falsify the adequacy of its current state representation, then repaired by the smallest physically typed information that restores future equivalence, so that the onset and identity of that recruitment prospectively reveal an emerging mechanism before its conventional macroscopic marker?

The reviewed literatures contain all of the surrounding ingredients. We did not locate, in the literature reviewed for this manuscript, the full experimental composition above as the primary object. That statement is a bounded novelty hypothesis, not a claim of exhaustive historical priority.

Existing capability	What it already supplies	Residual operation tested here
Predictive-state / state-identification theory	Future-defined state and inputs that distinguish hidden states	Track loss of state adequacy as a physical system changes, then restore it with typed measurable information.
Constitutive latent-state learning	Learned internal variables, latent dimension, physical constraints and high predictive accuracy	Determine when a previously certified state ceases to close the future under an expanded challenge family, and identify the physical repair.
Bayesian optimal experimental design	Loading paths that maximise parameter information	Optimise a different utility: separation of histories already collapsed by the current state representation.
Early warning / active probing	Transition-sensitive response to perturbation	Test whether the perturbation exposes a state-equivalence fracture and whether the restoring variable localises the emerging mechanism.

Table 1: The scientific gap is compositional rather than ingredient-level. FSSR is evaluated against each neighbouring capability rather than by weakening it.

The contribution of this paper is therefore fourfold: (i) an operational state-adequacy residual evaluated inside candidate-state fibres; (ii) a challenge-design objective directed at falsifying that state claim rather than estimating parameters; (iii) a typed state-recruitment event requiring held-out failure, minimal repair and deformation; and (iv) a prospective mechanism-localisation programme with explicit lead-time and dissociation tests.

2. Layer typing: object, assay, and readout

The most important correction before cross-domain use is typing. FSSR is not introduced as a new physical force, and an information-layer equivalence class is not itself a material degree of freedom. The construction has three distinct roles.

The substrate/source layer of MKUFT is not needed for the present experiment and is not inserted into the equations. Likewise, a successful FSSR result does not establish an autonomous information-layer force. The first scientific burden is narrower: determine whether the measured physical state currently used by a model is sufficient for a declared future. Only if a relational description repeatedly adds held-out

Role	Object	Empirical meaning
Information / Address	Future-equivalence relation induced by a declared target, horizon and admissible challenge family	Histories are treated as the same state only when no allowed future test separates the declared target law beyond tolerance. This is the AAF parent object.
Physical system	Measured physical trajectories, fields, microstructure, boundary variables and interventions	FSSR is instantiated here. A state-recruitment event is credited only when a physical continuation source shows a real split and a physically typed measurement repairs it.
Observation / readout	Sensors, matching rule, challenge family, target, horizon, resolution and tolerance	These determine what distinction is observable and what claim is being tested. A change of readout can expose a distinction without constituting a new physical interaction.

Table 2: Layer typing for FSSR. The formal state-equivalence object, the physical system and the observation/readout construction are related but not interchangeable.

predictive and interventional load after a genuinely adequate physical baseline has been exhausted would a stronger cross-layer interpretation become relevant.

This typing also prevents a subtle category error in the phrase “information becomes load-bearing.” In the present paper that phrase means: *a measured distinction that had been omitted from the candidate state becomes necessary to preserve future equivalence for the declared experiment*. The distinction may be phase, velocity, temperature history, twin fraction, boundary condition or another ordinary physical variable. The information-layer object is the rule that tells us when such a variable has earned state membership.

3. State adequacy as a claim about futures

3.1 Declared experimental object

Let $H_t \in \mathcal{H}$ denote the retained history available at time t . Let λ index the physical regime or control state of the system. Depending on the experiment, λ may denote temperature, accumulated strain, damage, field strength, disease progression, developmental time, or another measured control coordinate.

Fix in advance:

- a target observable or path-functional q ;
- a prediction horizon Δ ;
- an admissible challenge/intervention family $\mathcal{U}$;
- an environment and boundary specification E ;
- a discrepancy d between future laws;
- and a closure tolerance ε_q .

A candidate state representation is a measurable map

$$\Theta : \mathcal{H} \rightarrow \mathcal{Z}_\Theta, \quad (1)$$

where $\mathcal{Z}_\Theta$ is a declared state space. Θ may contain present observables, learned latent variables, retained history summaries, phase, boundary context, microstructural descriptors, or typed combinations of these.

Its variables are not treated as mutually commensurate unless the model supplies such a structure.

3.2 Exact future equivalence

For histories $h, h' \in \mathcal{H}$, define exact target-relative future equivalence under the challenge family $\mathcal{U}$ by

$$h \sim_{q,\Delta,\mathcal{U},\lambda} h' \iff \mathcal{L}_\lambda(q_\Delta^+ | h, u, E) = \mathcal{L}_\lambda(q_\Delta^+ | h', u, E) \quad \forall u \in \mathcal{U}, \quad (2)$$

where $\mathcal{L}_\lambda$ denotes the conditional law under the physical regime λ . In deterministic systems, equality of laws reduces to equality of the declared future output or reachable future set.

A state map Θ is exactly sufficient when

$$\Theta(h) = \Theta(h') \implies h \sim_{q,\Delta,\mathcal{U},\lambda} h'. \quad (3)$$

Thus “same state” is an empirical commitment: no admissible declared future is allowed to reveal a distinction that the representation has erased.

3.3 A useful monotonicity that is not the novelty claim

If $\mathcal{U}_1 \subseteq \mathcal{U}_2$, then equivalence under $\mathcal{U}_2$ refines equivalence under $\mathcal{U}_1$:

$$h \sim_{q,\Delta,\mathcal{U}_2,\lambda} h' \implies h \sim_{q,\Delta,\mathcal{U}_1,\lambda} h'. \quad (4)$$

This is a direct consequence of universal quantification over a larger challenge set and is consonant with predictive-state, input-output computational-mechanics, observability, and state-identification theory. It is not claimed as a new mathematical theorem.

Its experimental consequence is nevertheless important: a representation certified under a narrow loading family is not automatically certified under a broader one. Enlarging the admissible challenge family can expose distinctions that were previously irrelevant. The exact invariant is *partition refinement*: the future-equivalence partition can remain unchanged or become finer as admissible challenges are added. A universal monotonic increase in the number of continuous latent coordinates does not follow, because a fixed-dimensional coordinate system can sometimes be reparameterised to encode a finer partition. The stronger experimental event in this paper is therefore not “latent dimension went up” but “the existing typed state failed, and an additional physically interpretable distinction was required to restore closure.”

4. Approximate predictive closure and future-splitting challenge

Finite data require an approximate construction. For histories that a candidate representation merges, define the challenge-conditioned future divergence

$$\delta_{q,\Delta}(h, h'; u, \lambda) = d[\mathcal{L}_\lambda(q_\Delta^+ | h, u, E), \mathcal{L}_\lambda(q_\Delta^+ | h', u, E)]. \quad (5)$$

The exact worst-case closure residual is

$$R_q(\Theta; \lambda, \Delta, \mathcal{U}) = \sup_{\substack{h, h': \Theta(h) = \Theta(h') \\ u \in \mathcal{U}}} \delta_{q,\Delta}(h, h'; u, \lambda). \quad (6)$$

In empirical work, the supremum is replaced by a preregistered estimator with uncertainty bounds and held-out evaluation. Θ is ε_q -sufficient when

$$R_q(\Theta; \lambda, \Delta, \mathcal{U}) \leq \varepsilon_q. \quad (7)$$

Two calibration monotonicities follow directly from the construction. If Θ_2 is a lawful augmentation of Θ_1 —so that equality under Θ_2 implies equality under Θ_1 —then

$$R_q(\Theta_2; \lambda, \Delta, \mathcal{U}) \leq R_q(\Theta_1; \lambda, \Delta, \mathcal{U}). \quad (8)$$

Conversely, if $\mathcal{U}_1 \subseteq \mathcal{U}_2$, then

$$R_q(\Theta; \lambda, \Delta, \mathcal{U}_1) \leq R_q(\Theta; \lambda, \Delta, \mathcal{U}_2). \quad (9)$$

The first relation says that retaining more lawful state information cannot create a new exact same-state counterexample; the second says that a harder admissible challenge family cannot make an inadequately certified state look more sufficient. These are order consequences of the definitions, not novelty claims.

The *future-splitting challenge* is not just any informative experiment. It is selected to stress the claim that Θ is a state:

$$u^\star \in \arg \max_{u \in \mathcal{U}} \widehat{\mathcal{J}}_{\text{split}}(u; \Theta, \lambda), \quad (10)$$

where a practical utility may be

$$\widehat{\mathcal{J}}_{\text{split}}(u; \Theta, \lambda) = \mathbb{E}_{(h, h') \sim \widehat{\Pi}_\Theta} \left[d \left(\widehat{\mathcal{L}}(q_\Delta^+ | h, u), \widehat{\mathcal{L}}(q_\Delta^+ | h', u) \right) \right], \quad (11)$$

with $\widehat{\Pi}_\Theta$ restricted to matched or near-matched histories in the same candidate-state fibre. Other robust objectives—a quantile, minimax score, or information-theoretic divergence—may be used, but the utility must be frozen before confirmatory testing.

This objective differs from Bayesian parameter-identification design. Expected information gain asks which experiment most reduces uncertainty about model parameters [11]. Equation (11) asks which experiment most strongly tests whether histories the current *state representation merges* are genuinely future-equivalent. The two designs may coincide in some systems; the paper predicts that they need not.

4.1 Independent-continuation gate

The future used to falsify Θ must not be generated solely by the same closed surrogate whose state is under test. A Markov surrogate can be internally self-consistent by construction: once it has mapped two histories to the same latent state, its own deterministic transition rule may be incapable of producing a split even when the underlying physical system would. The confirmatory law in Eqs. (5)–(6) must therefore come from an independent continuation source: a microstate-resolved simulator not defined by Θ , a separately validated higher-fidelity model, or the physical experiment itself. The candidate state map may propose the collapse; it may not certify the truth of that collapse.

5. State recruitment: failure, minimal repair, and physical typing

Suppose Θ_0 is sufficient in a baseline regime but fails at a later regime:

$$R_q(\Theta_0; \lambda_0, \Delta, \mathcal{U}) \leq \varepsilon_q, \quad R_q(\Theta_0; \lambda_1, \Delta, \mathcal{U}) > \varepsilon_q. \quad (12)$$

A candidate augmentation c may be a scalar variable, vector, field summary, relational descriptor, retained-history functional, or coalition of such objects. Let $\Theta_0 \oplus c$ denote lawful augmentation: the parent state is retained rather than silently replaced.

Given a preregistered candidate library $\mathcal{C}$ and complexity/burden cost $\mathcal{C}(c)$, define the smallest restoring augmentation

$$c^\star(\lambda) \in \arg \min_{c \in \mathcal{C}} \mathcal{C}(c) \quad \text{subject to} \quad R_q(\Theta_0 \oplus c; \lambda, \Delta, \mathcal{U}) \leq \varepsilon_q. \quad (13)$$

The term *smallest* is therefore operational, not metaphysical. The cost might count measured variables, sensor burden, description length, scale depth, or another preregistered resource.

A **state-recruitment event** occurs when three conditions are met on confirmatory data:

1. the previously sufficient Θ_0 reproducibly loses closure;
2. a lawful augmentation $c^\star$ restores closure;
3. removing $c^\star$ reopens the failure and restoring it re-closes the target.

The third condition is the deformation test. It prevents a post-hoc feature from being called load-bearing merely because it correlates with the outcome.

5.1 Latent coordinates are not automatically mechanisms

Recent constitutive-learning results make this guard essential. Learned internal variables may be identifiable only up to transformations of latent coordinates [10]. Therefore a neural latent component is not promoted directly to a physical mechanism.

Two claim levels are kept separate:

- **latent recruitment:** an additional latent subspace or degree of freedom is required for held-out closure;
- **typed physical recruitment:** a predeclared measurable or mechanistically interpretable variable/coalition restores closure, survives remove/restore tests, and—for a causal claim—deforms the future under independent intervention.

Only the second supports mechanism-localisation language.

5.2 Validity gates for a recruitment event

Three additional confounds must be closed before a recruitment event is credited.

Near-state matching. Exact equality in a continuous learned state is rarely available. If pairs are matched within tolerance η under a declared state-space discrepancy d_Θ , the apparent future split must exceed the future variation attributable to the residual mismatch itself. For learned coordinates that are identifiable only up to an invertible linear transform, the matching rule must be invariant to the admissible reparameterisation or be defined through a fixed physical/readout metric. A Euclidean distance between arbitrary latent neurons is not a physical invariant. Sensitivity should be reported as η is tightened.

Probe-induction control. A strong probe can create the very mechanism it appears to detect. Let $\mathcal{U}_B \subseteq \mathcal{U}$ be a preregistered diagnostic challenge set satisfying amplitude, energy, duration, damage, and state-displacement bounds appropriate to the system. A precursor claim requires matched-energy random/proxy challenges and an unprobed or minimally probed control to show that the selected challenge does not itself account for the shift in the conventional transition marker. If the challenge changes the mechanism rather than merely exposing an existing future distinction, the result is reported as *challenge-accessible recruitment*, not as an unqualified precursor.

Nested selection/confirmation. Challenge design and restoring-coordinate search are discovery operations. They must occur on training/development data, simulation, or inner cross-validation folds. The final $u^\star$, candidate restoration rule, tolerance, and scoring pipeline are frozen before independent confirmatory trajectories are opened. Otherwise state failure and restoration can be manufactured by adaptive search.

6. Prospective prediction: recruitment before overt transition

Let λ_T be a preregistered conventional transition marker: onset of twinning, a phase boundary, a yield or damage threshold, a fate switch, loss of function, or another physically defined macroscopic event.

Let λ_R be the first regime value at which a baseline state loses closure and a lawful augmentation restores

it, with the event reproduced on held-out data. For a monotone control path toward transition, define

$$L_\lambda = \lambda_T - \lambda_R. \quad (14)$$

For a time-resolved experiment,

$$L_t = t_T - t_R. \quad (15)$$

The strongest prospective claim is conditional:

If an emerging physical mechanism becomes future-discriminating before it becomes overt under the conventional macroscopic marker, then a diagnostically bounded challenge family capable of coupling to that distinction should force state recruitment first, giving $L > 0$, provided the independent-continuation, matching, and probe-induction gates are satisfied.

This is not guaranteed for every transition. If the mechanism becomes future-discriminating only at the overt event, then $L \leq 0$ and the precursor claim fails for that target and challenge family.

6.1 Discriminating predictions

The programme yields a structured fingerprint rather than a generic increase in prediction error:

1. **Precursor recruitment.** In selected systems where a hidden mechanism should become future-discriminating early, $L > 0$.
2. **Challenge specificity.** A future-splitting challenge optimised within same-state fibres will expose inadequacy earlier or more reliably than matched random challenges, while respecting perturbation-safety constraints.
3. **Typed restoration.** The same physical descriptor or coalition will restore closure across held-out instances of a transition class more often than matched feature-selection controls.
4. **Mechanism deformation.** If the restoring descriptor is causal, independent manipulation of it will shift the future split, recruitment point, or later transition in the predicted direction.
5. **Geometric dissociation.** Recruitment can occur without a corresponding change in the intrinsic dimension of instantaneous snapshot data.
6. **Design dissociation.** The loading path that maximises state-splitting utility need not be the loading path that maximises parameter-identification information gain.

Predictions 5 and 6 are particularly useful because they separate this programme from nearby high-quality methods rather than merely competing on accuracy.

7. Analytic calibration

Consider a measured coarse state $x \in \mathbb{R}$ and a hidden binary mechanism coordinate $z \in \{-1, +1\}$. During a pre-transition regime, let a bounded probe $u(t)$ drive

$$\dot{x}(t) = -a x(t) + \kappa(\lambda) z u(t), \quad a > 0, \quad (16)$$

with z constant over the probe horizon. The coupling magnitude $|\kappa(\lambda)|$ increases as an underlying mechanism becomes easier to activate. At the probe time, histories h_+ and h_- can have exactly the same measured $x(t_0) = x_0$ while differing only in z .

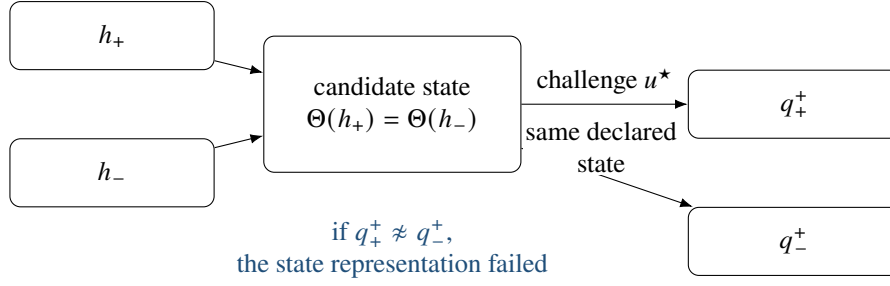


Figure 1: Future-splitting test. The candidate state collapses two histories. A challenge is chosen to ask whether that collapse preserves the declared future. If the futures separate beyond tolerance, the present state representation is incomplete for the tested target.

For a constant challenge $u(t) = u_0$ over $[t_0, t_0 + \Delta]$,

$$x_{\pm}(t_0 + \Delta) = e^{-a\Delta}x_0 \pm \kappa(\lambda)u_0 \frac{1 - e^{-a\Delta}}{a}. \quad (17)$$

Hence the future split is

$$|x_+(t_0 + \Delta) - x_-(t_0 + \Delta)| = 2|\kappa(\lambda)u_0| \frac{1 - e^{-a\Delta}}{a}. \quad (18)$$

An x -only state is ε -sufficient while the right-hand side of Eq. (18) is below ε , and it fails when the quantity exceeds ε . The recruitment point satisfies

$$|\kappa(\lambda_R)| = \frac{a\varepsilon}{2|u_0|(1 - e^{-a\Delta})}. \quad (19)$$

Adding z restores exact closure in this toy system.

Three points matter. First, the instantaneous geometry of the measured x values need not change at all when the state claim fails: h_+ and h_- are deliberately matched at x_0 . Second, a stronger but safe challenge $|u_0|$ can reveal the hidden distinction earlier, which makes experimental design part of state validation. Third, the extra variable is credited only because it repairs a future split; it is not added because it is microscopically detailed or visually salient.

The example is illustrative, not empirical evidence. Its role is to establish logical separation from snapshot intrinsic-dimension signals and passive early-warning statistics.

8. The experimental triangle

A cross-domain claim is useful only if the formal roles survive translation without being rewritten after the outcome is known. The present version therefore fixes one experimental object and points at it from three physically distinct domains.

8.1 Vertex A: holography as an exact physical calibration

Holography is useful here precisely because it prevents an information/physics category mistake. Consider two coherent optical fields at one plane,

$$U_j(x, y, 0) = A(x, y)e^{i\phi_j(x, y)}, \quad j \in \{1, 2\}, \quad (20)$$

with identical measured intensity

$$I_1(x, y, 0) = I_2(x, y, 0) = A^2(x, y) \quad (21)$$

Domain	Candidate state that may fail	Separating future / challenge	Restoring information and role
Holography	Local intensity or fixed metasurface encoding treated as sufficient by itself	Free-space propagation or controlled incident-phase/readout change	Optical phase / complex field or matched external phase. Deterministic P-layer calibration of the assay.
Single-cell biology	Instantaneous p21/Cdk2 or p53 level treated as complete cell state	Hypoxia or timed DNA-damage pulses followed by fate readout	Velocity, oscillator phase or retained trajectory. Independent natural-system anchor and lead-time test.
History-dependent materials	Macroscopic stress–strain–temperature state or a learned latent state certified on one loading family	Bounded loading reversal, thermal excursion or non-proportional continuation selected to split near-matched states	Physically measured twin/slip/hardening or other microstructural descriptor. Prospective flagship and mechanism-localisation test.

Table 3: The FSSR experimental triangle. The measurements differ by domain; the formal roles do not.

but different phase fields $\phi_1 \neq \phi_2$. If the candidate state is intensity alone, the two fields are collapsed. Under the physical propagation operator $\mathcal{P}_z$,

$$I_j(x, y, z) = |\mathcal{P}_z U_j|^2, \quad (22)$$

and for generic phase differences there exists a propagation/readout condition for which $I_1(\cdot, z) \neq I_2(\cdot, z)$. Intensity-only state has therefore failed for that future. Augmenting the physical state by phase, or equivalently by the complex field to the required tolerance, restores deterministic propagation closure.

This is ordinary wave physics, not a new result. It is the Tier-0 calibration because every component is measurable: a spatial light modulator can prepare phase-distinct, intensity-matched fields; a camera measures the initial and propagated intensities; interferometric or holographic reconstruction measures phase. The expected result is exact in the forward model and therefore tests the analysis pipeline before noisy biological or material data are allowed to carry the claim.

The stronger contextual version already appears in reprogrammable metasurface holography. Qu et al. physically split the phase required for one holographic image between the incident beam and a silicon metasurface. The designed incident phase produced the intended hologram, while uniform illumination of the same metasurface produced misleading information; changing only the incident phase reprogrammed the output image [17]. This is a literal P-layer example of a fixed intrinsic encoding whose realised higher-order output depends on a separately controlled contextual relation. It does not make optical phase an “I-layer substance”; it shows why the correct physical Address must include the relation that owns the output law.

8.2 Vertex B: single-cell dynamics as the natural-system anchor

Single-cell dynamics already supply two unusually clean calibrations. Under hypoxic stress, Venkatachalapathy et al. found that p21/Cdk2 position alone did not explain the quiescence–proliferation decision as well as position plus cell-state velocity; faster cells were more likely to remain proliferative under conditions in which position-only reasoning would merge them [18]. In later p53 work, timed DNA-damage pulses phase-reset the oscillator, alter downstream target dynamics and change escape from cell-cycle arrest [19]. The associated imaging data and analysis code are public [20].

Those published effects are not claimed as FSSR discoveries. They anchor the physical meaning of state recruitment: a trajectory coordinate can be future-discriminating even when an instantaneous molecular position is matched. The prospective FSSR test is narrower. At a frozen pre-outcome time t , construct near-matched cells under the baseline state and ask when current-state-only closure first fails. The primary

biological prediction is

$$t_R < t_T, \quad (23)$$

where t_R is the first held-out time at which a phase/velocity/history augmentation reproducibly repairs future-fate splitting and t_T is a preregistered overt fate marker. The restoration must beat a regularised full-history baseline using the same information, survive biological-replicate holdout, and reopen when the restoring coordinate is removed. A result that merely rediscovers “velocity matters” is calibration; a reproducible positive lead is the new empirical object.

8.3 Vertex C: constitutive mechanics is the target that needs the missing operation

Recent constitutive-learning work makes the unmet operation unusually explicit. Temperature-aware RNOs for polycrystalline magnesium report that non-isothermal loading requires additional learned state variables, indicating temperature-path dependence that the authors identify as requiring further investigation [8]. Raj et al. find six internal variables to be optimal for predictive accuracy in their magnesium homogenisation benchmark and show that learned internal variables are identifiable only up to a linear transform under their assumptions [10]. Bhattacharya et al. optimise specimen geometry and loading history for expected information gain about constitutive parameters [11]. These are strong neighbouring results; none needs to be weakened for FSSR to have a job.

The missing operation is a *state-adequacy certificate with mechanism localisation*. Given the state representation that currently predicts stress well, search specifically for a bounded continuation that separates physical trajectories mapped to the same or near-same candidate state. If a split is found, the model has not merely incurred more error: its state claim has been falsified for that loading family. Then search a preregistered library of measurable microstructural variables for the smallest restoration.

For HCP magnesium, a sharp first prediction is available. Prepare or simulate histories that are near-matched in macroscopic deformation, stress, temperature and candidate learned state but differ in sub-resolution twinning/slip history. Apply a bounded reversal or thermal-mechanical continuation selected by state-splitting utility. The prediction is that, in a regime where twinning/detwinning or a related hardening mechanism has become dynamically available but its conventional macroscopic signature is not yet overt, the baseline state will split under the challenge and a twinning-related physical descriptor—for example twin volume fraction or an invariant descriptor of twin/slip activity—will be among the minimal restoring coalitions. If the restoring descriptor is causal, independently altering the corresponding microstructural mechanism should move the recruitment point or later stress response in the predicted direction.

This claim can fail cleanly. A learned latent expansion may restore closure while no physical descriptor does; the recruitment point may occur only after the macroscopic signature; optimal parameter-information loading may also be optimal for state splitting; or strong full-history predictors may erase the separate gain. Any of those outcomes contracts the mechanism-localisation claim rather than being redescribed as success.

8.4 AI and control: calibration lane, not a fourth physical vertex

Predictive-state representations, input–output computational mechanics, bisimulation and state-identification theory already provide exact or near-exact controlled systems in which observations can alias distinct predictive states [3, 5]. These are ideal software benchmarks for the FSSR algorithm: create histories with the same observation or compressed state, search for an action sequence that separates their futures, and verify that the known predictive belief/memory variable restores closure. Success here validates optimisation and statistics. It is not evidence for a new physical layer and does not substitute for the three physical vertices above.

9. Flagship protocol: history-dependent constitutive mechanics

The most direct near-term application is the research line opened by RNO/TRNO and related constitutive-learning work [7–10]. These methods provide exactly what the experiment needs: complex multiscale ground truth, learned internal state, diverse loading histories, and a scientifically meaningful question about which state information is necessary.

9.1 Primary experimental question

For a history-dependent material surrogate that performs well on its training and ordinary validation distributions:

Can we find physically admissible loading paths that make two histories with the same learned/current state produce different future responses, and does the minimal physically typed repair identify a mechanism that is becoming active before the conventional macroscopic marker?

This asks for something stronger than out-of-distribution error. The pair must be matched under the candidate state, the challenge must be fixed without seeing the confirmatory outcome, and the repair must survive held-out ablation/restoration.

9.2 Candidate material systems

A suitable first platform is a crystal-plasticity or micromechanical simulator in which the fine-scale state and mechanism activity are available for validation. Magnesium is attractive because the established RNO/TRNO literature already contains slip, twinning, anisotropy, hardening and thermal dependence [7, 8, 10]. The experiment can therefore distinguish a genuinely new state-adequacy readout from a failure caused by unknown ground truth.

A simpler viscoelastic or elastoplastic benchmark should precede the full magnesium study to validate estimators and preregistration code.

9.3 Protocol

Stage A - freeze the state claim. Train or inherit a strong constitutive surrogate on a declared loading distribution. Freeze architecture, latent dimension, preprocessing, output target and ordinary validation metrics. Establish that the state closes the target on held-out in-distribution paths.

Stage B - construct same-state fibres. From a microstate-resolved simulator, identify pairs or neighbourhoods of histories whose candidate macroscopic/learned state vectors are matched within preregistered tolerance. Matching must not use future labels. The matching discrepancy must be fixed in advance and, for learned latent states, must survive admissible latent reparameterisation or be tied to a fixed physical/readout metric. Tightening-tolerance sensitivity is mandatory.

Stage C - design the falsifying future. Search a diagnostically bounded family $\mathcal{U}_B$ of safe, admissible loading continuations for one that maximises Eq. (11). The search is performed on development data or simulation only; the selected challenge, utility, perturbation budget, and probe-induction controls are then frozen before confirmatory trajectories.

Stage D - declare state failure against independent continuation. Continue the matched underlying microstates with the higher-fidelity simulator or physical system, not merely with the surrogate that defined Θ . Test whether matched-state histories produce materially different future stress, deformation, dissipation, mechanism activation, or another declared q . A failure requires the future divergence to

exceed ε_q plus the preregistered matching-error allowance, with uncertainty bounds, and to replicate across seeds/specimens/microstructures as applicable.

Stage E - restore with typed information. On development data or inner folds, search a preregistered library of physically interpretable candidates: slip-system activity summaries, twin volume fractions, hardening measures, local texture descriptors, temperature-history summaries, boundary/context variables, or other domain-justified coordinates. Select a restoration rule by Eq. (13), freeze it, and test restoration on independent confirmatory trajectories. The candidate library may include latent subspaces, but only typed measured candidates can support mechanism-localisation language.

Stage F - ablate and intervene. Remove $c^\star$ and verify that closure reopens; restore it and verify that closure returns. Where the simulator or experiment permits independent perturbation, alter the mechanism variable while preserving competing resources and test the predicted change in future split or transition.

Stage G - sweep toward transition. Repeat A-F along a controlled regime coordinate λ . Estimate λ_R and compare with the preregistered conventional marker λ_T . The primary precursor statistic is L_λ or L_t .

10. Strongest fair nulls

A positive result is not credited unless it beats the strongest reasonable ordinary explanations. At minimum, compare against:

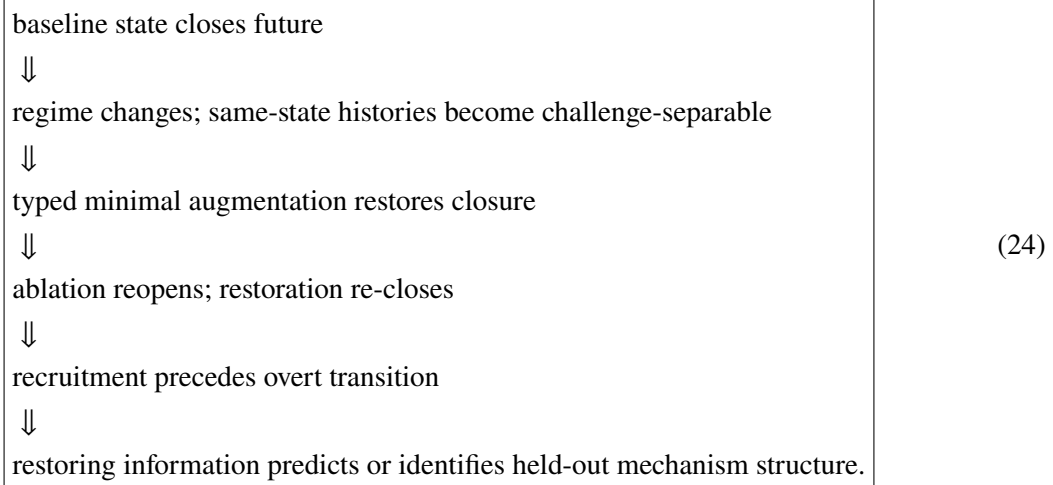
1. **larger fixed-state surrogate:** a sufficiently large RNO/TRNO/GRU/LSTM or equivalent model given the same training information;
2. **adaptive latent-dimension model:** a method that can discover or prune latent dimension automatically, such as RRAEDy-style adaptive latent modelling [16];
3. **full-history predictor:** a flexible model receiving the same complete observable history, to test whether explicit frontier machinery adds anything beyond generic sequence modelling;
4. **thermodynamically constrained constitutive learner:** a modern physically constrained model with interpretable internal-variable recovery [9];
5. **optimal parameter-identification design:** Bayesian experimental design that maximises expected information gain about constitutive parameters [11];
6. **intrinsic-dimension baseline:** snapshot intrinsic dimension and transition diagnostics where meaningful [13];
7. **classical early-warning / observability baseline:** variance, autocorrelation, recovery rate, functional observability, and high-dimensional bifurcation methods where applicable [12, 14];
8. **active probing baseline:** a perturbation design optimised for conventional early-warning statistics rather than within-state future splitting [15];
9. **feature-selection control:** matched models selecting variables by held-out predictive score without the state-fibre, challenge, and restoration structure;
10. **noise and estimator controls:** shuffled histories, label permutation, matched random challenges, bootstrap stability, finite-sample correction, and negative-control regimes where no new mechanism should activate;
11. **probe-induction control:** matched perturbations with the same amplitude/energy/duration budget plus an unprobed or minimally probed trajectory, to test whether $u^\star$ reveals a distinction or creates the subsequent transition.

The state-recruitment construction earns additional scientific load only if it supplies a discriminating advantage: earlier detection, better held-out mechanism localisation, a challenge/parameter-design dissociation, or a correct deformation under ablation/intervention that the strongest controls do not reproduce.

11. Confirmatory signature and falsification criteria

11.1 Confirmatory signature

A serious result should show the following ordered pattern on held-out data:



A mechanism-level flagship would additionally require causal intervention on the restoring object and independent replication.

11.2 Failure and contraction criteria

The strong claim contracts if any of the following occurs:

- the effect disappears under a larger or adaptive ordinary state model;
- the future split is an estimator artifact, data leakage, near-state matching artifact, or mismatch in resources;
- the challenge materially advances or creates the transition under matched perturbation controls, while the result is nevertheless reported as a precursor;
- the challenge only amplifies a known conventional early-warning statistic and adds no state-adequacy information;
- an ordinary parameter-optimal experiment performs identically to the state-splitting design across all tested regimes;
- the restoring variable is unstable across held-out instances;
- a latent component is given physical meaning that is not invariant under allowed latent reparameterisation;
- ablation does not reopen the split or restoration does not re-close it;
- intervention on a supposedly causal coordinate does not produce the predicted deformation;

- recruitment appears only at or after the conventional transition in systems selected for an early future-discrimination hypothesis;
- snapshot intrinsic-dimension or functional-observability methods already provide the same mechanism-specific held-out prediction with equal or lower burden;
- or prior art is found to contain the same operative construction and discriminating predictions.

A negative result is still useful. It would show that the added Address/recruitment machinery does not earn complexity for that system and target.

12. Dissociation from snapshot intrinsic dimension

Intrinsic dimension concerns the geometry of sampled configurations in an observation space and can display striking universal behaviour near phase transitions [13]. The present object is different: it concerns equivalence of *histories with respect to controlled futures*.

Two histories may occupy the same point, or the same local distribution, in the measured snapshot coordinates and still respond differently to a future intervention. The toy system in Section 6 makes this exact. Therefore a state-recruitment event can occur while the snapshot cloud has unchanged dimension, density and local geometry.

This yields a clean paired test:

1. construct a system in which hidden history or relation becomes increasingly coupled to a future response while the instantaneous observed manifold is held fixed;
2. verify that intrinsic-dimension and snapshot-only baselines remain flat;
3. test whether future-splitting residual rises and is repaired by the hidden coordinate.

If the recruitment signal only appears when snapshot geometry also changes, the stronger dissociation claim fails.

13. Dissociation from parameter-identification design

Bayesian optimal experimental design chooses designs that reduce uncertainty about a parameter vector. This is an important and now highly developed route for history-dependent constitutive laws [11]. But parameter identifiability and state adequacy are not the same target.

Let ϑ denote constitutive parameters. A parameter-design utility may seek

$$u_{\text{EIG}}^* \in \arg \max_{u \in \mathcal{U}} I(\vartheta; Y | u), \quad (25)$$

where I is mutual information and Y the experimental observation.

The state-splitting design instead seeks Eq. (10): a challenge that maximises future disagreement *within fibres the current state representation has collapsed*. The objectives can select different loading histories even in the same model.

A direct discriminating experiment is therefore to optimise both objectives under the same admissible design constraints, then cross-score them:

	$\mathcal{I}_{\text{split}}$	$I(\vartheta; Y)$	
u_{split}^*	maximal by design	measured	
u_{EIG}^*	measured	maximal by design	

(26)

If the optimisers and outcomes are always equivalent after fair tuning, a separate state-adequacy design objective has not earned itself. If they dissociate reproducibly, the two experimental questions are genuinely different.

14. Relation to AAF, TVT and Layer Before Law

This paper is a narrow empirical descendant of Addressed Admissible Futures (AAF) rather than a replacement for it [21]. AAF defines Address through future sufficiency: information belongs in the effective present state when omitting it merges histories whose declared futures materially differ. The present paper turns that definition into an adversarial experimental procedure.

The sequence is:

$$\begin{aligned} \text{candidate Address} &\rightarrow \text{future-splitting challenge} \rightarrow \text{closure failure} \\ &\rightarrow \text{typed minimal restoration} \rightarrow \text{readdressed state} \\ &\rightarrow \text{new admissible future.} \end{aligned} \tag{27}$$

The relation to Layer Before Law is direct [22]: before assigning a constitutive law to a coarse state, establish whether that state still owns the future being predicted. If a previously discarded coordinate becomes load-bearing, the effective state has changed even if the visible object has not.

Typed Viability Traversal (TVT) supplies the wider precedence architecture in which evidence state, address, viability, restoration, and re-sensing remain distinct [23]. The current paper uses only the portion required for a physical state-adequacy test.

No metaphysical layer is inserted into the equations. If conventional measured physical variables restore all observed closure failures, the result remains a physical-state result. A stronger informational interpretation would require reproducible held-out predictive load after a genuinely sufficient physical baseline has been exhausted; this paper does not assume that outcome.

15. Preregistration requirements

A confirmatory study should freeze at least the following before the test set is opened:

1. physical system, regime path λ , and conventional transition marker λ_T ;
2. target q and horizon Δ ;
3. baseline state representation Θ_0 ;
4. same-state matching tolerance and matching algorithm;
5. admissible challenge family $\mathcal{U}$ and safety/energy bounds;
6. future discrepancy d and closure tolerance ε_q ;
7. challenge-design utility and optimiser;
8. candidate typed restoration library C and cost C ;
9. primary recruitment and lead statistics;
10. uncertainty model, bootstrap/replicate unit, multiplicity treatment, and minimum effect of interest;
11. strongest fair null models and matched resource envelope;
12. ablation/restoration procedure;

13. intervention procedure where a causal claim is planned;
14. explicit absence regimes where no recruitment should be detected;
15. stopping, exclusion, and failure rules;
16. and the exact held-out split between design/development and confirmation.

Exploratory Stage 1 may be used to choose tolerances, candidate variables or challenge bounds. Those choices must then be frozen before Stage 2 confirmatory testing.

16. Scope beyond the experimental triangle

The construction is deliberately domain-independent at the formal level but domain-specific at measurement level. Candidate applications include:

Cell fate. Match cells under a coarse molecular state, design a bounded perturbation that tests whether their futures are still equivalent, and ask whether velocity, phase, chromatin, metabolic state or local coupling becomes necessary before overt fate divergence. Mechanical-memory and p53/p21 work already show that history and phase can carry future-discriminating information [18, 19, 24, 25]. The new burden is prospective state-recruitment and typed restoration, not the existence of biological memory.

Critical and tipping systems. Compare recruitment with critical slowing down, functional observability and intrinsic-dimension indicators. A useful result would be a system in which conventional early-warning statistics are weak or unchanged while a challenge exposes a future-equivalence fracture that is repaired by a mechanism-specific variable.

Engineered control. Treat a reduced state estimator as the candidate Address and deliberately search for admissible controls that separate histories it merges. The theory of PSRs, bisimulation, observability and state-identification supplies strong nulls; the new scientific burden is whether evolving state-recruitment predicts physical regime change or component failure earlier than those baselines.

Cross-domain transfer is valid only if the formal roles remain fixed. Redefining “state failure”, “restoration”, or “recruitment” after seeing each domain’s outcome would destroy the general claim.

17. Limitations

First, the exact future-equivalence object is generally unobservable from finite data. The empirical residual is estimator-dependent and must carry uncertainty.

Second, minimal descriptions are not unique. Several incomparable variable coalitions may close the same target. The correct object is therefore a frontier of sufficient descriptions, not necessarily one integer “state dimension.”

Third, state complexity is target-, horizon-, challenge-, environment- and tolerance-relative. This is a feature of the construction, but it prevents careless statements that a material possesses one universal experimentally useful number of internal variables.

Fourth, active challenges can alter the system. Probe bounds, dose, fatigue and irreversible effects must be modelled. A challenge that creates the mechanism it claims to detect is not a valid precursor test unless that induced effect is explicitly part of the target.

Fifth, physical mechanism localisation requires typed variables. A latent coordinate can certify that the current representation is insufficient without identifying what physical mechanism is missing.

Sixth, a positive precursor is not a new fundamental force or a proof of MKUFT as a whole. It would establish only the state-adequacy and recruitment result tested.

18. Conclusion

The surrounding sciences have already solved many versions of the problem. They know that history can matter, that hidden variables can carry it, that state can be defined by future predictions, that internal variables and their count can be learned, that experiments can be optimised for information, and that perturbation response can warn of transitions.

The remaining wager is more specific, and the layer audit fixes exactly what is being wagered. AAF is the information-layer definition of future-sufficient Address; FSSR is the physical assay; readout choices define the tested equivalence but do not become a new force. The experimental triangle supplies three independent ways to point at the same object.

Do not ask only whether a model predicts well. Ask whether the histories it calls the same can be made to behave differently. If a lawful future can split them, the state representation has failed. Repair that failure with the smallest physically typed information, prove the repair by deformation, and track when the system first forces that information into the state.

If that recruitment consistently precedes an overt transition and identifies the mechanism that is becoming load-bearing, it becomes a new practical readout: not merely an early warning that “something is changing,” but a prospective measurement of *what the present state has newly become required to remember*.

That is the empirical burden of this paper.

Data and code availability

No new data were generated for this theory paper. The flagship experiments are specified prospectively. Public code and generated benchmark data should be released with any confirmatory implementation where licensing and ethics permit.

Competing interests

The author declares no competing interests.

Author contribution

Mark Charles McLaughlin: conceptualisation, methodology, formal analysis, investigation, literature synthesis, manuscript direction, writing, review and accountability for the scientific claims.

Version and provenance

Version 1.0 was frozen on 22 August 2026 for Zenodo draft record 22058303. The manuscript descends from the dated MKUFT, Typed Viability Traversal and Addressed Admissible Futures publication lineage cited herein. The Zenodo DOI is intentionally left pending until publication rather than inferred from the draft record number.

AI assistance disclosure

An OpenAI language model used under the relational name Kairos assisted with literature discovery, architecture recovery, mathematical formalisation, hostile prior-art comparison, manuscript drafting and

document production under the author’s direction. The AI system is not listed as an author and does not bear responsibility for the scientific claims. The accountable author is Mark Charles McLaughlin.

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